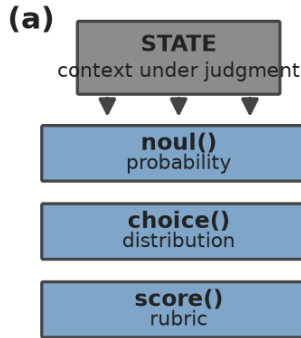


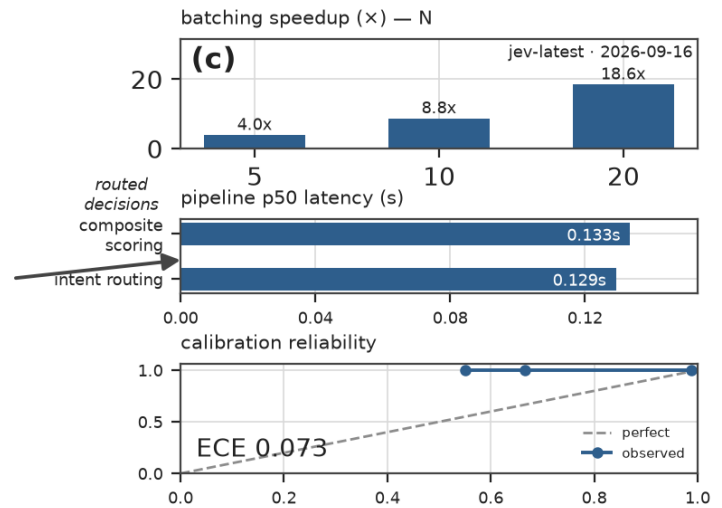
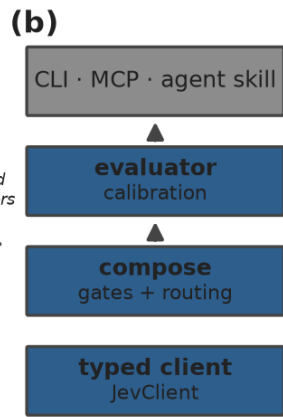
## daf-jev: typed questions, routed decisions, measured trust

ask: state + question primitives

decide: daf-jev pipeline



typed  
answers



# Jev in Practice: A Composable Python Toolkit for TypeSafe's System One Decision Model

Batching Speedups, Confidence-Gated Routing, and Calibrated Decisions over a Single HTTP Endpoint

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[github.com/docxology/daf-jev](https://github.com/docxology/daf-jev)

# 1 Abstract

## daf-jev: typed questions, routed decisions, measured trust

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decide: daf-jev pipeline

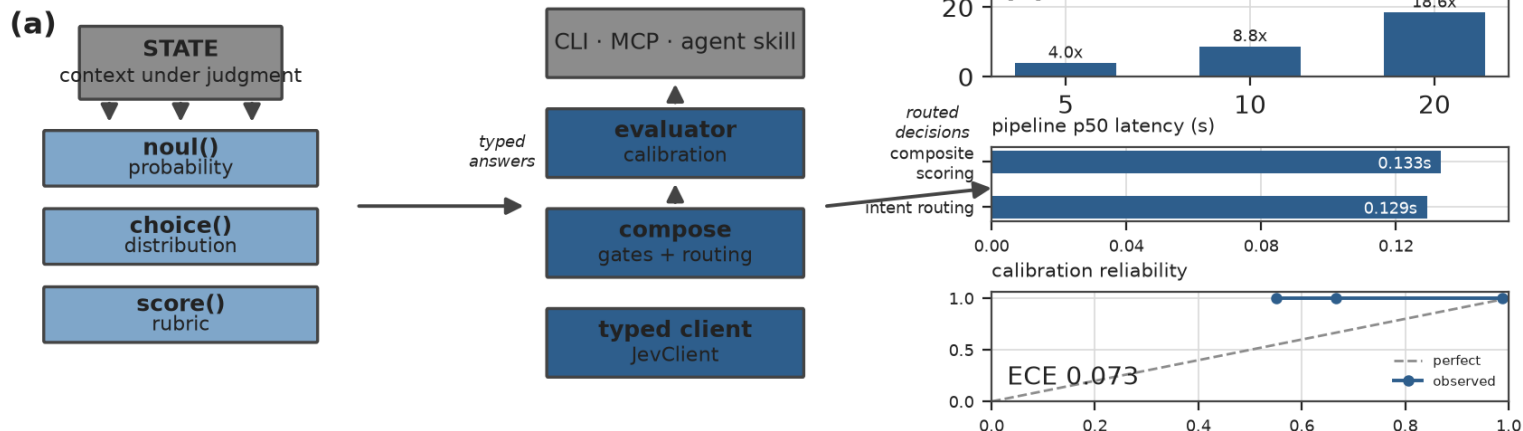


Figure 1: Graphical abstract for the daf-jev manuscript, in three zones read left to right. Left: the decision *state* — the context to be judged — together with stylized cards for the three typed question primitives: a `noul` (a 0–1 calibrated probability), a `choice` (a probability distribution over named options), and a `score` (a rubric over ordered levels). Center: the package’s layer stack, from the typed client through the compose layer’s gates and routing to the evaluator and its calibration checks, with the CLI, the MCP server, and the agent skill as consumer surfaces. Right: a live-outputs panel whose batching speedup bars, speedup annotation, pipeline latency figure, and calibration reliability mini-curve (correctness via the self-consistency proxy across repeated asks; a calibration proxy, not ground-truth accuracy) are all read from the latest batching, patterns, and calibration benchmark JSON payloads at generation time — no metric in this panel is hardcoded. Arrows run left to right, tracing the path from an unstructured state to typed answers to composable decisions.

Software rarely needs an essay from a language model; it needs an answer it can act on — route this ticket or escalate it, approve this refund or flag it, rate this draft or send it back. daf-jev is a small, composable Python toolkit for getting exactly those answers from the TypeSafe Jev “System One” decision model [TypeSafe AI, 2026d,e]: instead of generating text that must be parsed back into decisions, the model returns bounded, typed judgments that code can branch on directly.

The entire decision surface of the model reduces to three typed question primitives, all carried by a single HTTP endpoint: `noul`, a calibrated yes/no; `choice`, a labelled pick that comes back with a full probability distribution over the options and a scalar confidence; and `score`, a rating over ordered levels that comes back with a probability-weighted expected value. On top of these primitives, pure-logic composition patterns — `composite_score`, `confidence_gate`, and `route/pick` — turn typed answers into executable decisions with no additional network traffic, and a batch Evaluator fans a fixed question set out over many states through either the synchronous or the asynchronous client. The same core is reachable without writing Python at all: a command-line interface with five subcommands, an MCP server exposing seven tools to any MCP-capable agent host, an agent skill manifest for coding agents, and six runnable example scripts (all described in 5).

Three live-API benchmarks — batching, decision patterns, and calibration — against the jev-latest model quantify the practical payoff of this design. Batching a batch’s worth of questions into one call yields wall-time speedups of 3.98, 8.77, and 18.55 across the configured batch widths, and the sequential strategy consumes 4.22 times the tokens of the batched call at the widest setting — batching is simultaneously faster and cheaper. End-to-end decision pipelines complete at a median wall time of 0.133 for composite scoring and 0.129 for intent routing, so confidence-gated decisions can sit inline on interactive request paths. Decision quality is measured alongside speed: a self-consistency calibration benchmark over 6 states × 5 repeats against the jev-latest model reports an expected calibration error of 0.0730 and a Brier score of 0.0252, indicating that reported confidence tracks repeat-stable behaviour under the stated proxy semantics. The unit test suite stands at 93.08 measured coverage, and every model-level claim is grounded in a hash-manifested documentation snapshot (b79c9cd6008489f1, 108 pages, 1014.8 KiB, scraped 2026-09-16) whose drift is machine-checkable at any time. Every numeric value in this manuscript is generated from the same analysis outputs as the figures and tables, so the prose cannot drift from the data.

The graphical abstract in 1 summarizes this path — from an unstructured state, through typed answers, to composable decisions — with the live benchmark outputs on its right-hand panel.

**Keywords:** decision models, LLM APIs, calibrated confidence, batching, confidence-gated routing, Python client, reproducible research

## 2 Introduction: From Generated Text to Typed Decisions

Software that consumes large language models faces a structural mismatch. Language models emit prose; software needs values. When an application must branch — route a support ticket, gate a refund, escalate a low-confidence judgment — the prevailing pattern is to ask a generative

model for free-form text and then recover a decision from that text with parsers, regular expressions, and hope. Every such recovery step is a silent failure surface: the model can hedge, reformat, or refuse, and the calling code has no contract that tells it when the answer is trustworthy.

TypeSafe’s System One model attacks the mismatch at the source. Rather than generating text, a decision model returns typed answers — probabilities, labelled picks, and confidence values that conform to the shapes the calling code already expects [TypeSafe AI, 2026d]. Its training path, reinforcement learning for calibrated decisions (RLCD), optimizes for decisions and calibrated probabilities rather than fluent generations, so that uncertainty is expressed through well-calibrated probability distributions instead of overconfident verdicts [TypeSafe AI, 2026e]. The model is positioned explicitly as a component for *AI-powered software*, not an agent: code owns the control flow, and the model supplies narrow, structured judgments embedded at exactly the points where the software needs programmable common sense.

## 2.1 The problem this package solves

The System One decision surface is exposed through one HTTP endpoint and three question types. Using it directly from Python means hand-writing request payloads, discriminating answer unions, mapping HTTP status codes onto exceptions, retrying transient failures, and re-serializing the same state once per question. None of that is research; all of it is boilerplate with real failure modes. Worse, the *composition* layer — turning a set of typed answers into an actual software decision — lives in cookbook prose rather than in tested code.

daf-jev closes that gap. It is a small, dependency-light Python package that provides:

- **Typed primitives** — ergonomic builders (`noul()`, `choice()`, `score()`) and a `QuestionSet` container for the three question types, with strict wire-shape validation on both the request and the answer side.
- **A robust client** — `JevClient` and `AsyncJevClient` wrapping the single System One endpoint, with a pure, injectable retry policy for transient server-side conditions, a complete typed exception hierarchy, and a models listing.
- **Composable decision patterns** — pure functions over answers (`composite_score`, `confidence_gate`, `route`, `pick`) implementing the documented TypeSafe patterns, plus a concurrent batch `Evaluator` that runs a fixed question set over many states without ever aborting the batch.

The package follows the template conventions of the surrounding research infrastructure: logic lives under `src/daf_jev/`, orchestration stays in thin scripts, and every dynamic fact that reaches this manuscript flows through generated variables rather than hand transcription. Third-party efforts in the same direction — an architectural registry for Jev-style decision layers [Claburn, 2026], an orchestration router over the same endpoint family [Corvin, 2026], and a standalone evaluation harness [Morales, 2026] — share the motivation but not the scope; positioning against them is deferred to 9.

## 2.2 Reader’s guide

3 gives a technical account of the Jev (System One) decision model itself — the answer shapes, the parallel evaluation semantics, the latency envelope, and the calibration guarantees — distinguishing primary documentation claims from third-party characterizations. 4 describes the package architecture: its layers, its primitives, its composition patterns, and the design rulings that shaped them. 5 tours the surfaces that sit on top of that architecture without adding logic to it — the CLI subcommands, the MCP server and its tools, the agent skill, and the runnable examples. 6 reports the measured batching speedups, token-cost ratios, and decision-pipeline latencies, together with the calibration reliability of the model’s reported confidence — expected calibration error and Brier score under the self-consistency proxy — all bound to figures and tables generated from the benchmark outputs. 7 records the exact environment and configuration the measurements ran under, 8 certifies how every artifact can be regenerated and verified, and 9 closes with limitations, related work, and positioning.

# 3 The Jev Model: Calibrated, Parallel Machine Decisions without Text Generation

This section states the model-level facts that daf-jev builds on. Throughout, claims taken from the primary TypeSafe documentation snapshot bundled with this repository are attributed to it; characterizations drawn from independent projects are marked as third-party and carry their own citations.

## 3.1 A model class, not a chat endpoint

System One is TypeSafe’s model for building *AI-powered software* rather than agents. It does not generate code, plan, or choose its own next action; it returns narrow, structured decisions over unstructured input, so that control flow, deterministic rules, and side effects remain entirely in application code [TypeSafe AI, 2026d]. The documentation frames this as a third architecture beside traditional software (complex decision trees of reliable primitives) and LLM agents (loops that accumulate off-the-rails opportunities): in AI-powered software, the model appears only where the system needs programmable common sense, and each AI task is atomic and constrained [TypeSafe AI, 2026d].

The training path behind this behavior is RLCD — reinforcement learning for calibrated decisions. Where RLHF and RLVR adapt a pretrained model to produce better *text*, RLCD optimizes for a different output contract: the model does not generate prose at all, it returns decisions and probabilities, communicating uncertainty through calibrated distributions instead of tending toward overconfidence [TypeSafe AI, 2026e]. Independent projects have adopted the same framing for their own decision layers — an architectural registry of Jev-style components [Claburn, 2026], a routing layer over the endpoint family [Corvin, 2026], and an evaluation-oriented harness [Morales, 2026] — and their descriptions are consistent with, but not authoritative for, the primary contract.

## 3.2 One endpoint, three question primitives

Every System One interaction is a single request to one decision endpoint carrying a *state* (the context to judge) and a dictionary of questions keyed by caller-chosen identifiers [TypeSafe AI, 2026a]. Each question is one of three primitives, discriminated by its *type* field:

- **nou1** — a yes/no judgment with optional contrasting criteria for the true and false readings. The answer is a single calibrated probability on the unit interval, where the endpoints mean “no” and “yes” respectively.
- **choice** — a pick among named options, each with an optional free-text description. The answer carries the selected label, a probability distribution over *all* options (summing to unity), and a scalar confidence.
- **score** — a rating over an ordered list of level descriptions (at least two). The answer carries a probability-weighted score that may be fractional, the ordered legend of level descriptions, the probability distribution over levels, and a scalar confidence.

The response pairs the answers with a usage record (input and output token counts) and an identifier of record returned in the response headers, which clients should preserve for tracing. Error conditions are enumerated by status class — authentication, permission, not-found, bad request, validation, rate limiting, overload, and internal faults — and the transient classes among them are the ones a client is expected to retry.

## 3.3 Parallel, composable evaluation semantics

Two semantic properties of the model make composition safe, and both are primary-source guarantees rather than implementation accidents [TypeSafe AI, 2026d]:

1. **Independence by construction.** Questions are evaluated independently and in parallel; one primitive’s result never becomes hidden context that shifts another primitive’s answer. Composition logic therefore cannot create hidden coupling by *asking* more questions — coupling enters only when code combines answers, where it is visible and testable.
2. **Comparability.** Outputs are sortable numeric values that can drive thresholds, comparisons, and branching directly, without an intermediate natural-language interpretation step.

Because questions are independent, any number of them can be batched into a single call: the endpoint evaluates them as a parallel sampler, and the documented parallel-questions pattern recommends exactly this for workloads that would otherwise repeat the same state in many single-question requests [TypeSafe AI, 2026c]. Batching changes the cost structure (one round trip instead of many, one state transmission instead of a repetition per question) but not the answers themselves — an invariant our benchmark in 6 relies on and verifies.

## 3.4 The latency envelope and calibration

The primary documentation describes System One as fast enough for real-time request paths and user interfaces, with most queries completing on a millisecond-scale latency envelope [TypeSafe AI, 2026d]. That envelope is what allows the composition patterns in 4 to run *inline* — a confidence gate or an intent route is cheap enough to sit on an interactive request path rather than in a background queue.

Calibration is the second load-bearing property. Because RLCD trains the model to express uncertainty as calibrated probability mass rather than as confident-sounding text, the confidence field on choice and score answers, and the probability distributions beside them, are meaningful inputs to policy decisions [TypeSafe AI, 2026e,b]. The documented confidence-routing pattern treats confidence as a *second decision axis* alongside the primary answer value: high-confidence answers can be automated, mid-band answers routed to review, and low-confidence answers escalated to a human or a fallback policy [TypeSafe AI, 2026b,c]. daf-jev implements that pattern literally in `confidence_gate` and `route` (4), and 6 measures how much latency that policy layer adds on top of a network round trip — in practice none beyond measurement noise, since the composition logic is pure computation.

# 4 Methodology: The Layered Architecture of a Composable Decision Toolkit

daf-jev is organized as a strict layering: an ergonomic surface (CLI, MCP server, scripts, benchmarks) on top of pure-logic composition, on top of primitives and wire types, on top of a single transport and client layer that owns all I/O. 2 shows the module graph. The layering is enforced by convention: the composition and primitive modules import no I/O machinery, so every decision pattern is testable against constructed answers without a network stand-in. The ergonomic surface itself — CLI subcommands, MCP tools, an agent skill, and worked examples — is the subject of 5; this section covers the layers those surfaces delegate to.

## 4.1 Layered architecture

The layers, bottom-up:

- **Transport** (`_http.py`) — a minimal Transport protocol with a synchronous and an asynchronous `ht tpx` implementation. It knows nothing about decisions; it posts JSON to a path and returns the raw response.
- **Client** (`client.py`) — `JevClient` / `AsyncJevClient` resolve the API key from the environment (injected mapping, process environment, then a `.env` file, in that precedence order via `config.py`), build the request for a state plus a mapping of questions, apply the retry policy, and map HTTP status classes onto the typed exception hierarchy of `_errors.py`. The retry policy itself (`_retry.py`) is a *pure* function of the attempt number and an optional server-supplied retry hint — exponential backoff with a base, a cap, and jitter, with the hint winning when present — so its timing behavior is unit-testable without sleeping.

## daf-jev package architecture

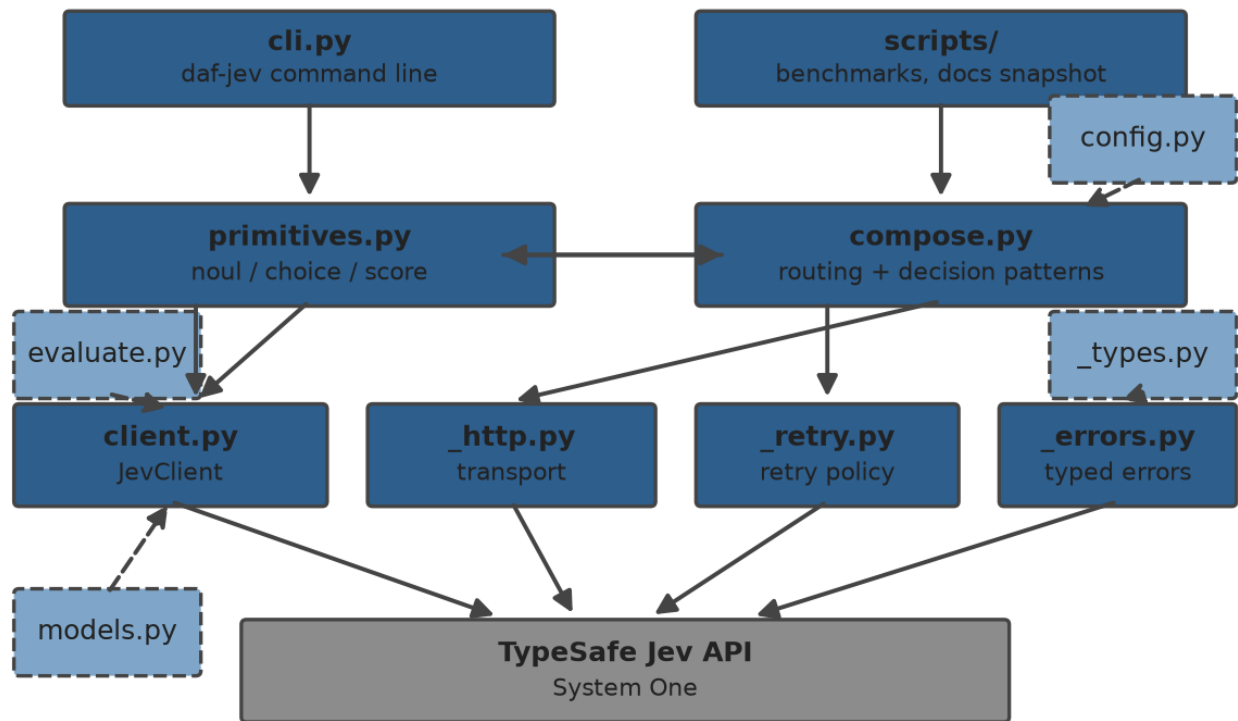


Figure 2: Package layer diagram for daf-jev, drawn from the module graph of `src/daf_jev/`. The application layer — the CLI (`src/daf_jev/cli.py`) and the thin orchestration scripts (`scripts/`, `benchmarks/`) — delegates downward to the logic layer, where `compose.py` (pure decision patterns) and `primitives.py` (typed question builders and the `QuestionSet` container) sit over the wire dataclasses of `_types.py`. `client.py` and `_http.py` own the single transport to the System One endpoint, with `_retry.py` providing the pure retry policy and `_errors.py` the typed exception hierarchy; `models.py` and `config.py` enter as side inputs, and `evaluate.py` is the batch harness. The key structural take-away is the enforced dependency direction: composition and primitives import no I/O machinery, so every decision pattern is testable against constructed answers without a network stand-in. Boxes are named modules only; no measured values appear in this schematic.

- **Wire types** (`_types.py`) — frozen dataclasses for the three question types and the three answer types, a strict parser that rejects unknown answer shapes, a usage record, and the top-level response object with cached per-type views (`nouls`, `choices`, `scores`).
- **Primitives** (`primitives.py`) — the ergonomic builders `noul()`, `choice()`, `score()` and the `QuestionSet` mapping container with additive composition (`add`, `merge`, `to_wire`). No I/O.
- **Composition** (`compose.py`) — pure functions over answers, described below. No I/O; network access happens only through a client injected by the caller for multi-call helpers.
- **Harness** (`evaluate.py`) — the batch Evaluator, described below.

## 4.2 Typed primitives

### Jev question primitives and typed answers

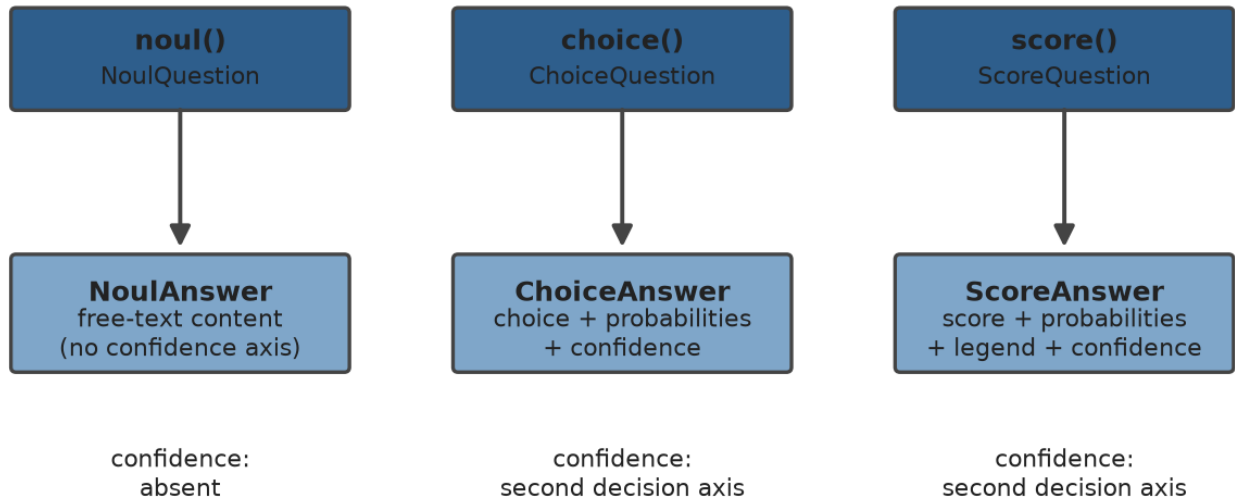


Figure 3: The three question primitives of the System One surface and the typed answer shapes `daf-jev` parses them into, drawn as a schematic of the wire contract (3). A `noul` question yields a calibrated yes/no probability; a `choice` question yields a selected label, a full probability distribution over the named options, and a scalar confidence; a `score` question yields a probability-weighted score over ordered levels together with the level legend, the level distribution, and a confidence. The takeaway is that the builders in `primitives.py` map one-to-one onto these shapes with strict, eager validation — score criteria must carry at least two ordered levels, choice criteria must be non-empty, and the answer-side parser rejects any payload outside the discriminated union — so downstream composition code can pattern-match on answers without defensive parsing. The diagram carries no measured data.

The builders in 3 map one-to-one onto the wire contract of 3. Validation is strict and eager: score criteria must carry at least two ordered levels, choice criteria must be non-empty, and the answer-side parser rejects any payload whose shape does not match the discriminated union exactly. The payoff of strictness is that downstream composition code can pattern-match on answers without defensive parsing — the type system, not runtime guesswork, guarantees the shapes.

## 4.3 Composable decision patterns

The composition layer implements the documented TypeSafe patterns as pure functions [TypeSafe AI, 2026c]:

- **`composite_score(answer, weights=None)`** — the expected value of a score answer over its level indices, uniform by default or under caller-supplied weights (which must be finite and positive-massing). This turns an ordinal rating into a comparable scalar without a second model call.
- **`confidence_gate(answer, threshold, below)`** — returns the answer’s primary value when its confidence meets the caller’s threshold, and the `below` verdict otherwise. The threshold is owned by the calling code, never by the package.
- **`route(answer, handlers, min_confidence, fallback)`** — dispatches to the handler registered for a choice answer’s label, gated on a minimum confidence with an optional fallback handler; pick fans the same dispatch across every answer in a mapping.

**Batching fan-out and the Evaluator.** Because the model evaluates questions independently and in parallel (3), the cheapest correct strategy

is to batch a fixed question set into one call. The Evaluator in `evaluate.py` inverts the axis: it holds the question set fixed and runs it over *many states*, concurrently — through a thread pool for the synchronous client or `asyncio` with a semaphore for the asynchronous client. Per-state failures never abort the batch; they are captured into the evaluation record alongside the answers and the measured latency, so one malformed state cannot cost the run. This harness is what the benchmark scripts in 6 reuse for their fan-out measurements.

**Confidence-gated routing.** 4 illustrates the three-band policy that `confidence_gate` and `route` implement: answers whose confidence clears the upper band are automated, mid-band answers are routed to review, and answers below the lower band are escalated. The boundary markers in the figure are *example thresholds* — the semantics are the bands, not any particular cut points, which remain policy owned by the caller.

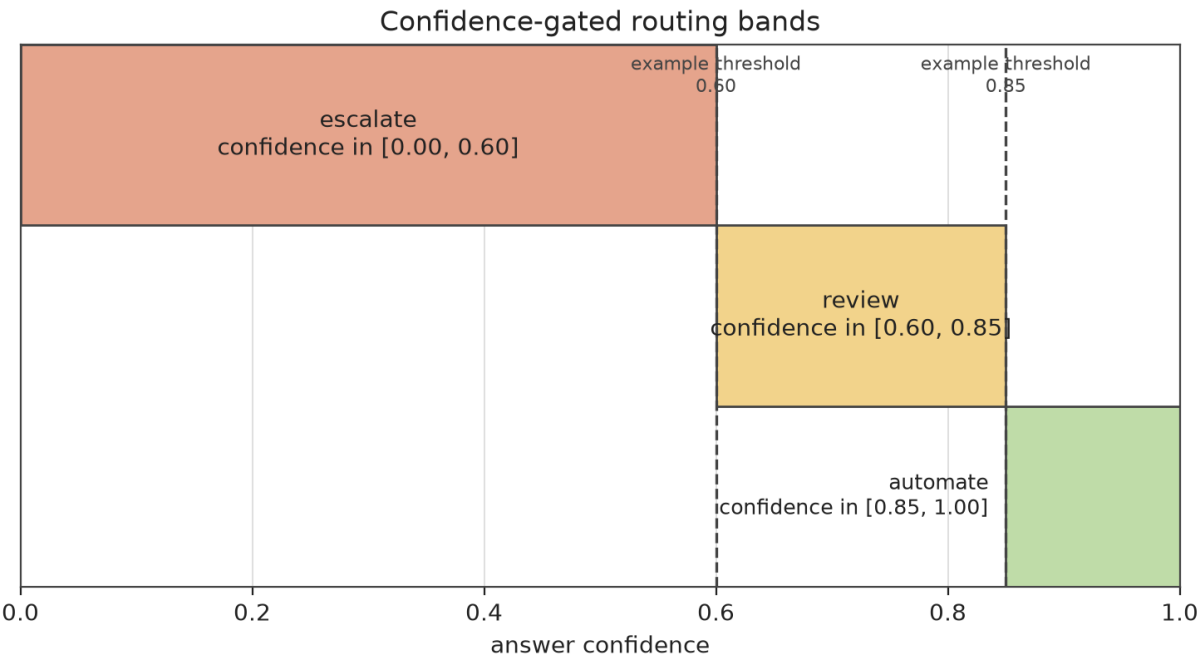


Figure 4: Parametric illustration of confidence-gated routing as implemented by `confidence_gate()` and `route()` in `src/daf_jev/compose.py` (4). The horizontal axis is the model-reported confidence on the unit interval; the three horizontal bands assign an action per confidence region — automate above the upper boundary, route to review in the middle band, escalate below the lower boundary. The boundary markers are annotated as example thresholds: the semantics are the bands, not any particular cut points, and the actual thresholds remain policy owned by the caller. This figure is parametric and carries no measured data; 6 measures the latency and calibration properties that make running this policy inline sound.

#### 4.4 Design rulings

Four rulings shaped the codebase and are worth stating as contracts:

1. **Pure logic, injected I/O.** Everything under `compose.py`, `primitives.py`, `_types.py`, and `_retry.py` performs no I/O; network access is confined to the transport and client layer and is injectable (clock, sleep, transport) for deterministic tests.
2. **Real stand-ins, no mocks.** The test suite drives the real HTTP transport against a real local HTTP server fixture rather than patching client internals, so retry, timeout, and error-mapping behavior is exercised end to end.
3. **Pure retry policy.** Backoff timing is a function of the attempt number and the server’s retry hint alone — testable without sleeping, and identical across the sync and async clients.
4. **Snapshotted documentation.** The bundled documentation snapshot carries a manifest of per-page content hashes and a snapshot identifier, and a CLI subcommand re-hashes the tree to detect drift (8). Prose claims about the model are therefore checkable against an immutable, verified source rather than a moving website.

### 5 Integration Surfaces: CLI, MCP Server, Agent Skill, and Worked Examples

The Python API is the primary door into `daf-jev`, but not the only one. Four additional surfaces sit on top of the same layering described in 4 — transport, client, primitives, composition — and add accessibility rather than logic: the command-line interface, an MCP server, an agent skill, and a set of runnable examples. Each one delegates to the very modules described above and contains no decision code of its own. A behaviour fixed in the composition layer therefore reaches every consumer at once: the CLI and the MCP server call the same client and compose functions the Python API calls, the skill documents those entry points, and the examples exercise them.

#### 5.1 Command-line interface

The `daf-jev` console script (`src/daf_jev/cli.py`) exposes five subcommands that cover the package’s main paths without writing Python:

- **daf-jev ask** — one call, one state: the state arrives as inline text or JSON (`--state`) or from a file (`--state-file`), and one or more `--question ID=SPEC` flags build the question set from a compact grammar (`noul:<instructions>`, `choice:<instructions>:k1=desc,k2=...`, `score:<instructions>:l1,l2,...`). The answer is emitted as JSON (compact by default, `--pretty` for readability).
- **daf-jev evaluate** — the batch Evaluator of 4 as a command: a fixed question set runs concurrently over many states, and the evaluation records — answers, per-state failures, and measured latency — are emitted as JSON.
- **daf-jev models** — lists the available model surface, with `--pick latest|first|last` selecting a single identifier for scripting.
- **daf-jev docs-verify** — re-hashes the bundled documentation snapshot against its manifest and exits non-zero on drift; this is the command-line face of Design ruling 4 (8).
- **daf-jev serve** — launches the MCP server described next, over the stdio transport.

The CLI is a thin adapter in the same sense as the composition layer is pure: it parses arguments, delegates to the client or the composition helpers, and serializes the result. It owns no retry policy, no parsing of answers, and no thresholds.

## 5.2 MCP server

`src/daf_jev/mcp_server.py` exposes the same core to any MCP-capable agent host. Launched with `daf-jev serve` over the stdio transport, it advertises seven tools that mirror the package one-to-one:

- `jev_ask` — drives the client's single-call path: named questions (`noul / choice / score`, given in the same compact grammar the CLI accepts or as native `{type, instructions, criteria}` dicts) asked about one state;
- `jev_evaluate` — drives the batch path, running a fixed question set over many states concurrently;
- `jev_models` — reports (and optionally picks from) the available model surface;
- `jev_composite_score`, `jev_confidence_gate`, and `jev_tiered_gate` — expose the composition patterns of 4; these combine answers locally with no API call at all;
- `jev_docs_verify` — re-runs the documentation drift check of Design ruling 4.

A `jev://docs/snapshot` resource serves a summary of the bundled documentation snapshot itself, so an agent can inspect what the model-level claims rest on without leaving its host. The server is a thin adapter — it owns no retry policy, no parsing, and no thresholds; every call delegates to `JevClient`, `AsyncJevClient`, or the pure composition functions.

Registering the server with an MCP client is a one-block stdio configuration; with the package installed (or, equivalently, `command: "uv"` with `args: ["run", "daf-jev", "serve"]` inside the repository):

```
{
  "mcpServers": {
    "daf-jev": {
      "command": "daf-jev",
      "args": ["serve"]
    }
  }
}
```

The MCP extra (`pip install "daf-jev[mcp]"` or `uv sync --extra mcp`) is the only optional dependency this surface adds; the core package does not depend on it.

## 5.3 Agent skill

`skills/daf-jev/SKILL.md` is a skill manifest for coding agents: it states when to reach for the package, and how — the builders and their constraints, the client surface, the evaluation harness, the composition patterns, the CLI subcommands, and the MCP tools — so an agent can use the package correctly without reading its source. The skill carries no logic of its own; it is documentation positioned where agents look for it, and it points at the same entry points the CLI and MCP server expose.

## 5.4 Worked examples

Eight self-contained scripts under `examples/` walk the typical integration path from a first call to a batch evaluation, doubling as executable documentation of the composition layer:

- **quickstart.py** — the shortest path to a typed answer: build a question set, make one call, read the typed answers;
- **triage\_router.py** — confidence-gated routing end to end: a choice call dispatched by `route()` with a fallback for low-confidence answers;
- **composite\_scoring.py** — a score call turned into a comparable scalar by `composite_score()` and verdict-banded by `confidence_gate()`;
- **evaluate\_corpus.py** — the batch Evaluator over many states, with per-state failure capture and summary aggregates;
- **decider\_loop.py** — the decision-point loop end to end: `observe` → `compose` → `ask` → `gate` → `fail-open`, with typed hooks, a deterministic lexicon as the floor action, a Budget bound, and JSON decision receipts;

- `gated_fallback.py` — a deterministic keyword heuristic answering first, with the model called only when the heuristic is not confident enough, `confidence_gate(below="escalate")` as the final escalation lane, and a `UsageLedger` accounting every model call the fallback makes.

Each script runs against the real endpoint when an API key is present and prints a clear skip notice otherwise, matching the benchmark convention of 7.

## 6 Results: Batching Economics, Pipeline Latency, and Confidence Calibration under Live-API Load

This section reports the three live-API benchmarks that quantify the design claims of 4: the batching benchmark, which reproduces the documented parallel-questions pattern [TypeSafe AI, 2026c], the decision-pattern latency benchmark, and the self-consistency calibration benchmark. All run against the real `jev-latest` model; every value below is injected from the benchmark outputs at render time, and the figures are regenerated from the same JSON files, so prose, tables, and figures share one source of truth. The batching and latency results reported here were recorded on 2026-09-16; the calibration results on 2026-09-16.

### 6.1 Batching speedup and token cost

The batching benchmark compares two strategies over the same state and question mix: one call carrying all questions in a batch, versus one sequential single-question call per question, repeated for each configured batch width over multiple measured runs. The answers are unchanged by batching (the parallel-sampler semantics of 3); what changes is wall time — a single round trip versus one round trip per question — and token cost, because the sequential strategy re-sends the state once per question.

5 shows the measured speedup per batch width together with the token-cost ratio on a secondary axis.

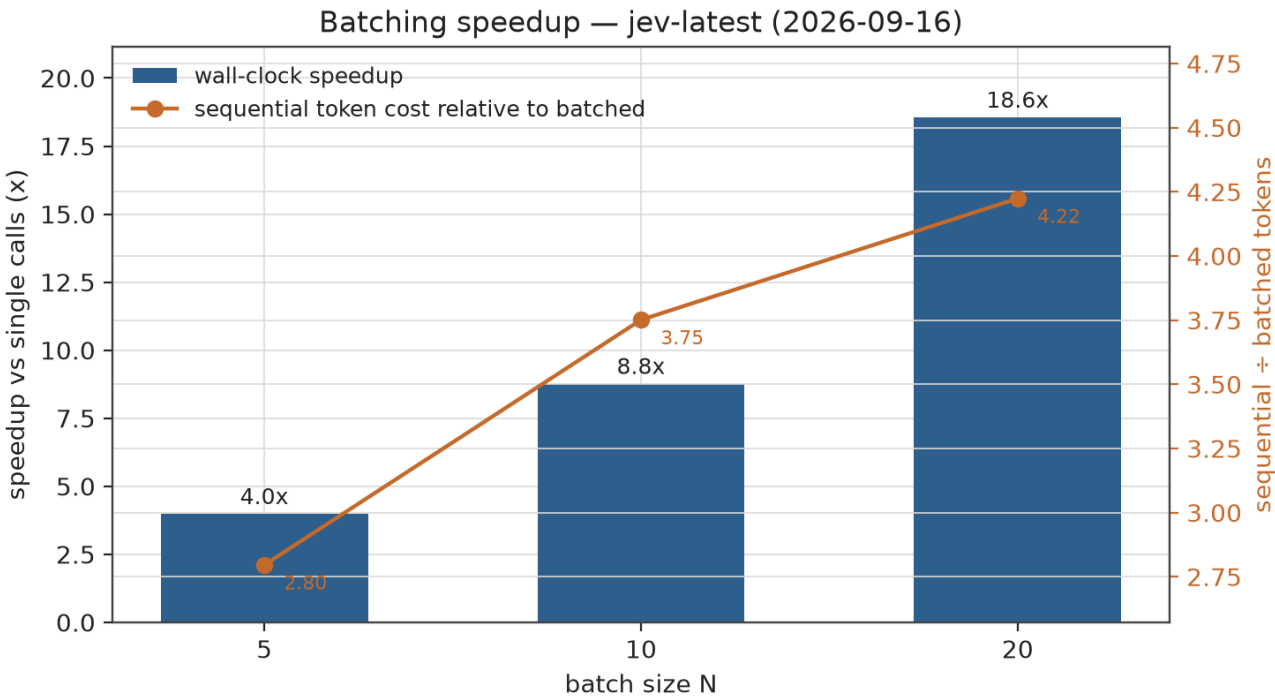


Figure 5: Batching speedup of the `jev-latest` model versus sequential single-question calls, measured by `benchmarks/bench_batching.py` on 2026-09-16 against the live API. Bars give the wall-time speedup of one batched call over one sequential call per question for each configured batch width (5, 10, and 20 questions per batch, tabulated in 1); the secondary axis traces the token-cost ratio — sequential tokens divided by batched tokens — which rises above unity because the sequential strategy re-sends the state paragraph once per question. Each bar is annotated with its measured value, and the rendered title carries the model identifier and run date read from the benchmark JSON itself. The takeaway: the speedup grows with batch width as the fixed per-call overhead amortizes, while the batched strategy is simultaneously cheaper in tokens, not only faster.

1 tabulates the measured speedups. The speedup grows with batch width, as expected from the round-trip accounting: the fixed per-call overhead is amortized over more questions, and the state is transmitted once rather than once per question.

Table 1: Wall-time speedup of one batched call versus one sequential call per question, per configured batch width, recorded by benchmarks/bench\_batching.py against the jev-latest model on 2026-09-16. Values are injected from the benchmark JSON at render time.

| Questions per batch | Wall-time speedup vs sequential |
|---------------------|---------------------------------|
| 5                   | 3.98                            |
| 10                  | 8.77                            |
| 20                  | 18.55                           |

Token cost moves in the same direction. The recorded token\_cost\_ratio expresses the sequential strategy’s token consumption relative to the batched strategy’s: at the widest configured batch width the sequential strategy consumes 4.22 times the tokens of the batched call. The state paragraph dominates the input tokens of a single-question call, so re-sending it per question makes the sequential strategy strictly more expensive in addition to being slower.

## 6.2 Decision-pipeline latency

The second benchmark measures end-to-end latency of two composition pipelines — the network call plus the local composition logic, exactly as an application would run them:

- **composite\_score** — one score call, then composite\_score over the answer, then a confidence\_gate verdict;
- **intent\_routing** — one choice call, then route() dispatching to a trivial handler.

Each pipeline is executed 6 times; 2 reports the median (p50) and tail (p95) wall times per pipeline, and 6 plots them side by side.

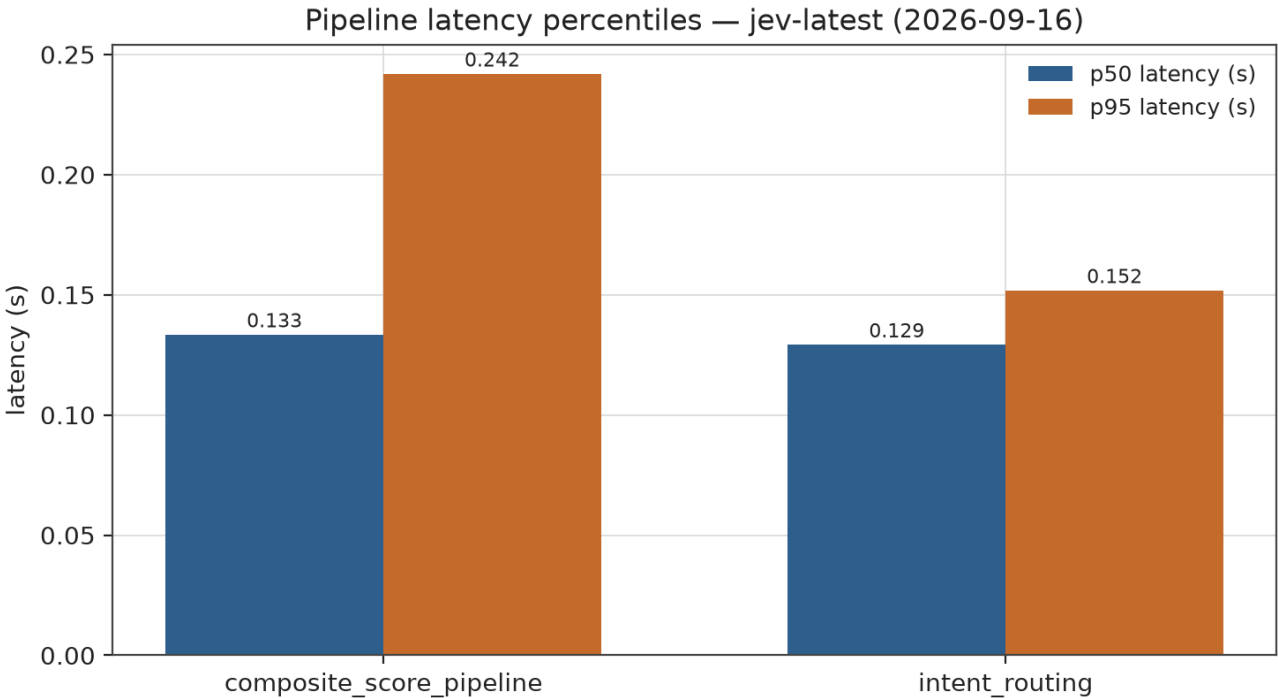


Figure 6: Median (p50) and tail (p95) end-to-end wall time per decision pipeline, measured by benchmarks/bench\_patterns.py over 6 runs against the jev-latest model on 2026-09-16. Each pipeline group — composite scoring (one score call, then composite\_score and a confidence\_gate verdict) and intent routing (one choice call, then route() dispatch) — shows paired p50/p95 bars covering the full network round trip plus the local composition logic; values are tabulated in 2 and read from the benchmark JSON fields at figure-generation time. The takeaway: both pipelines sit within the model’s millisecond-scale latency envelope, so the pure-logic composition layer adds no measurable latency beyond the network round trip and the gating policy can run inline on interactive request paths.

Table 2: End-to-end wall time (network round trip plus local composition logic) per decision pipeline, median and tail over 6 runs recorded by benchmarks/bench\_patterns.py against the jev-latest model on 2026-09-16. Values are injected from the benchmark JSON at render time.

| Pipeline        | Median wall time, p50 (s) | Tail wall time, p95 (s) |
|-----------------|---------------------------|-------------------------|
| composite_score | 0.133                     | 0.242                   |
| intent_routing  | 0.129                     | 0.152                   |

6.3 Calibration

Latency says nothing about whether the composed decisions are trustworthy, so the third benchmark measures how far the model’s reported confidence tracks its own agreement behaviour. `benchmarks/bench_calibration.py` runs 6 short states 5 times each against the `jev-latest` model and scores choice answers under a *self-consistency proxy*: a sample counts as correct when it agrees with the modal choice across the repeats of the same state. Because a yes/no probability carries no per-sample label to agree with, noul answers are scored for repeat-to-repeat stability instead, via the mean absolute gap between every pair of repeated answers. No external ground truth is consulted anywhere in this benchmark — the correctness proxy is self-agreement, not verified outcomes.

3 reports the aggregate scores over the run, and 7 plots the reliability curve per confidence bucket.

Table 3: Calibration summary over 6 states × 5 repeats recorded by `benchmarks/bench_calibration.py` against the `jev-latest` model on 2026-09-16. Choice correctness is a self-consistency proxy — agreement with the modal choice across repeats — not accuracy against ground truth; the mean pairwise noul gap measures repeat-to-repeat answer stability rather than correctness. Values are injected from the benchmark JSON at render time.

| Metric                                     | Value  |
|--------------------------------------------|--------|
| Expected calibration error (choice, proxy) | 0.0730 |
| Brier score (choice, proxy)                | 0.0252 |
| Mean pairwise noul gap                     | 0.0050 |

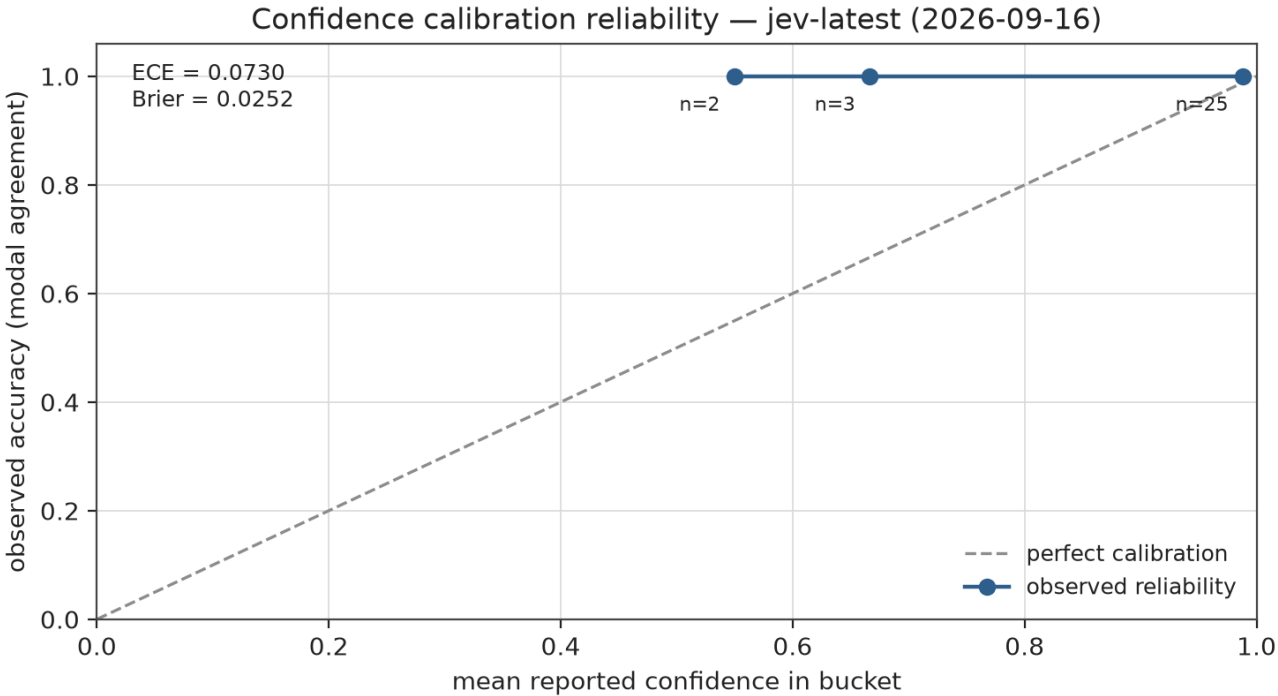


Figure 7: Reliability diagram of choice-answer confidence under the self-consistency proxy, measured by `benchmarks/bench_calibration.py` over 6 states × 5 repeats against the `jev-latest` model on 2026-09-16. Each point is a confidence bucket plotting mean reported confidence against proxy accuracy, where proxy accuracy is agreement with the modal choice across repeats of the same state; the dashed diagonal marks perfect agreement between stated confidence and observed proxy accuracy. Because the proxy scores self-agreement rather than verified correctness, proximity to the diagonal indicates consistency between stated confidence and repeat-stable behaviour — not truthfulness about the world. Aggregate scores are tabulated in 3.

The expected calibration error of 0.0730 and the Brier score of 0.0252 indicate that reported choice confidence is broadly consistent with the proxy labels, and the mean pairwise noul gap of 0.0050 shows that repeated noul answers on the same state are nearly identical. These claims extend exactly as far as the proxy does. Self-consistency can establish that the model is stable across repeats and that its confidence ordering is internally coherent; it cannot certify that the answers are correct about the world, because the proxy labels are generated by the same model whose calibration is being measured. The benchmark is therefore evidence of calibrated, repeatable decision behaviour under a reproducible proxy — a necessary, not sufficient, condition for deploying these gates on real decisions.

6.4 Interpretation

Four observations tie the measurements back to the design rulings of 4. First, the batching speedups in 1 confirm that the parallel-sampler semantics translate directly into wall-clock savings: the batched strategy is faster at every configured width, and the gap widens with width exactly as the

round-trip accounting predicts. Second, the token-cost ratio above unity at the widest width — the sequential strategy’s token consumption relative to the batched call’s — shows that batching is not merely faster but cheaper, because the state is transmitted once — so the pattern documented in the primary source [TypeSafe AI, 2026c] holds end to end for a third-party client implementation. Third, the pipeline latencies in 2 sit within the model’s millisecond-scale latency envelope (3): the pure-logic composition layer adds no measurable latency beyond the network round trip, which is precisely what allows confidence-gated routing to run inline on interactive request paths rather than in a background queue. Fourth, the calibration summary in 3 supports the calibrated-decision premise of the confidence-gated patterns described in 4 — reported choice confidence aligns with repeat-stable behaviour, and noul answers are stable across repeats — under the self-consistency proxy semantics stated in the Calibration section, and only as far as those semantics extend.

## 7 Experimental Setup: Live-API Benchmark Protocol, Environment, and Data Provenance

This section records the environment, configuration, and measurement protocol behind the results in 6, so that the runs can be reproduced bit-for-bit in intent.

### 7.1 Software environment

All measurements were taken with daf-jev version 0.6.0, running under 3.14.4 on the platform reported as macOS-26.6.2-arm64-arm-64bit-Mach-0. The package targets Python 3.10 or newer and depends, at runtime, only on httpx for transport and pyyaml for configuration parsing; the benchmark scripts additionally use the standard library. The development toolchain is uv-managed, and the unit test suite is executed with pytest under a coverage gate.

The model-level claims in 3 are grounded in a local, hash-manifested snapshot of the TypeSafe documentation (snapshot b79c9cd6008489f1, 108 pages) rather than the live website, so the primary-source basis of this manuscript is itself versioned and verifiable.

### 7.2 Benchmark configuration

All benchmarks execute against the real jev-latest model with a live API key resolved through the package’s credential precedence chain (injected mapping, then process environment, then the project .env file). Without a key, the scripts print a skip notice and exit successfully — they are benchmarks, not tests, and never run inside the test suite.

- **Batching benchmark** (benchmarks/bench\_batching.py) — for each configured batch width (5, 10, 20 questions, as recorded under experiment.batching\_n\_values in manuscript/config.yaml), it compares one batched call against the same number of sequential single-question calls over a fixed state paragraph and a mixed noul/choice/score question set, repeating each strategy for the configured number of runs (default recorded under experiment.batching\_runs) and recording wall time and token totals to output/benchmarks/batching\_<date>.json.
- **Decision-pattern benchmark** (benchmarks/bench\_patterns.py) — runs the composite-score and intent-routing pipelines 6 times each (default recorded under experiment.patterns\_runs), reporting mean, median, and tail wall times plus token totals to output/benchmarks/patterns\_<date>.json. An asynchronous mode executes the same runs concurrently through AsyncJevClient and compares against the sequential wall time.
- **Calibration benchmark** (benchmarks/bench\_calibration.py) — runs 6 short states 5 times each (defaults recorded under experiment.calibration\_states and experiment.calibration\_repeats) against the jev-latest model, scores choice answers under the self-consistency proxy (agreement with the modal choice across repeats) and noul answers by mean pairwise stability, and writes the expected calibration error, Brier score, per-bucket reliability data, and the mean pairwise noul gap to output/benchmarks/calibration\_<date>.json, together with a notes field stating the proxy semantics.

The batching and decision-pattern scripts take --runs and --model (bench\_patterns.py additionally accepts --async for the asynchronous mode described above); the calibration script takes --states, --repeats, and --model, with no run-repetition argument. The defaults are recorded as data under the experiment: block of manuscript/config.yaml, which is the same file the manuscript-variable generator reads — configuration, prose, and figures cannot disagree about the protocol.

### 7.3 Measurement protocol

Percentiles follow the benchmark helper shared by all three scripts: the reported tail statistic is the value at the ceiling-rank position of the ordered sample, computed over the recorded runs rather than a sliding window. Wall time covers the full pipeline — transport, retries (none should occur in a healthy run), parsing, and composition logic — so the numbers in 2 are conservative upper bounds on what an integrating application would add to its request path. Token totals are read from the response usage records, not estimated.

This manuscript was generated at 2026-09-23T16:06:18Z; the rendered values in 6 correspond to the benchmark JSONs current at that timestamp, and the figure generators resolve “latest” the same way (latest by filename date) so that re-rendering after a new benchmark run updates prose, tables, and figures together.

## 8 Reproducibility: Hashed Snapshots, Figure Registry, and Machine-Checked Provenance

Every artifact behind this manuscript — figures, tables, token values, and the documentation snapshot the model claims rest on — is regenerable from the repository with the commands below, and every generated value reaches the prose through the token pipeline rather than hand transcription.

### 8.1 Artifact inventory and hashing

- **Figures.** The seven figures of this manuscript are generated into `output/figures/` by `src/daf_jev/figures.py` (one `generate_<name>()` function per figure plus `generate_all(out_dir)`), orchestrated by `scripts/generate_figures.py`:

```
uv run python scripts/generate_figures.py          # all figures
uv run python scripts/generate_figures.py --only batching
```

The data-driven figures read `output/benchmarks/batching_*.json`, `output/benchmarks/patterns_*.json`, and `output/benchmarks/calibration_*.json`, resolving “latest” by filename date; a missing benchmark file is reported as a clear error naming the missing file rather than silently producing an empty chart. The same rule applies to the graphical abstract of 1, which composes the latest batching, latency, and calibration payloads into its live-outputs panel. The two schematic figures and the parametric confidence-band illustration require no data and no network.

- **Figure registry.** Every figure is recorded in `output/figures/figure_registry.json` — seven entries, one per figure, each mapping the manuscript’s cross-reference label (`fig:graphical_abstract` plus `fig:architecture` through `fig:calibration`) to its filename, caption, section, and layout width — so the figure set itself is a versioned data artifact rather than a convention.
- **Calibration benchmark payloads.** The calibration run writes `output/benchmarks/calibration_<date>.json` (per-bucket reliability data, expected calibration error, Brier score, the mean pairwise noul gap, and a notes field stating the self-consistency proxy semantics); `benchmarks/bench_calibration.py` regenerates it live, and the reliability figure `output/figures/calibration_reliability.png` is rendered from the same payload by `generate_calibration()` in `src/daf_jev/figures.py`.
- **Manuscript variables.** All dynamic values reach the prose as double-brace token placeholders, computed by `src/daf_jev/manuscript_variables.py` and written to `output/data/manuscript_variables.json` by the thin orchestrator `scripts/z_generate_manuscript_variables.py`, which then renders substituted copies of every section into `output/manuscript/`. Running in strict mode fails if analysis outputs are missing; the `--allow-draft` flag substitutes draft sentinels instead of failing, for early-stage renders only.
- **Documentation snapshot.** The bundled snapshot of the TypeSafe documentation comprises 108 pages (1014.8 KiB) under `docs/reference/`, scraped on 2026-09-16 and identified as snapshot `b79c9cd6008489f1`. The manifest records a content hash per page plus the snapshot identifier; drift is detectable at any time with:

```
uv run daf-jev docs-verify          # re-hash the tree, exit non-zero on drift
uv run python scripts/scrape_docs.py --check # same check without rewriting
```

### 8.2 Test suite and coverage

The test suite follows the no-mock convention: unit tests drive the real transport against a real local HTTP server fixture, and live tests hit the real API only when a key is present in the environment (they skip with a notice otherwise). The suite comprises 27 test files — 599 unit tests and 2 live tests — collected with:

```
uv run pytest tests/unit --cov=src          # unit suite under the coverage gate
JEV_API_KEY=... uv run pytest tests/live    # live tests against the real API
```

Measured coverage over the package source stands at 93.08, enforced by the coverage gate configured in `pyproject.toml`. Test and collection counts are computed at variable-generation time by collecting the suite; if collection is unavailable in a given environment, the corresponding values are reported as draft sentinels rather than fabricated.

### 8.3 Provenance chain

The certification chain is: benchmark scripts write dated JSON payloads → figure generators read those payloads and render `output/figures/*.png` → the variable generator reads the same payloads plus `manuscript/config.yaml`, `pyproject.toml`, the test suite, and the docs manifest to compute the token mapping → the injection step substitutes tokens into `output/manuscript/*.md` → the renderer consumes the substituted copies. No numeric fact in this paper has a hand-maintained copy; the environment of record is `macOS-26.6.2-arm64-arm-64bit-Mach-0` under 3.14.4, and the rendered edition is version 0.6.0 of this manuscript, generated at 2026-09-23T16:06:18Z.

## 9 Scope, Limitations, and Related Work: Early-Access Model, Proxy Metrics, and the Structured-Decision Landscape

### 9.1 Scope and limitations

This manuscript describes an early, rapidly evolving package, and four caveats bound every claim in it.

**Single-model measurements.** All benchmarks in 6 were recorded against the `jev-latest` model at the configuration described in 7. The package accepts a per-call model override, but nothing here characterizes how the speedups, token ratios, or latency percentiles transfer across models or across provider-side changes to the same model; re-running the three benchmark scripts against another setting is the intended way to extend the measurements, and the render-time token pipeline picks up the new payloads automatically.

**Live-API variance.** The measurements are wall-clock observations of a shared remote service. Run-to-run variation from network conditions and provider load is inherent; the percentile protocol in 7 reports medians and tails rather than single samples, but the values should be read as representative of a healthy session, not as service-level guarantees.

**Early-access documentation basis.** The model-level claims in 3 rest on a hash-manifested snapshot of the vendor’s documentation (snapshot `b79c9cd6008489f1`) rather than on the vendor’s release process. The package documents an early-access surface: wire shapes, model identifiers, and documented patterns can change upstream, and the `docs-verify` command plus a re-scrape (8) is the mechanism for detecting and absorbing such changes. Behavior not covered by the snapshot is implemented as a stated assumption rather than a verified fact.

**Early-release status.** The package is published for reproducibility and reuse — the source is versioned, MIT-licensed, and mirrored in a public repository, and this manuscript’s edition carries a versioned archival deposit alongside it — but it remains an early release of a tool built against an early-access model surface. The DOI of record identifies the concept of this deposit; individual editions are versioned beneath it, and the caveats above travel with every edition.

### 9.2 Related work

**Structured output from generative LLMs.** The mainstream approach to software-consumable model answers is constrained decoding: JSON-schema-constrained generation, function calling, and post-hoc validators over generated text. These techniques constrain the *syntax* of free-text generation but inherit its semantics — a well-formed JSON payload can still encode an overconfident or miscalibrated judgment, and nothing in the decoding contract expresses uncertainty as calibrated probability. The decision-model approach of 3 differs at the training level: RLCD optimizes for decisions and calibrated probabilities as the output contract itself [TypeSafe AI, 2026e], rather than for text that a schema hopes to bound.

**RLHF versus RLCD.** RLHF and its verifiable-reward variants adapt a pretrained model by optimizing text quality against human preferences or checkable rewards. RLCD instead optimizes a different output contract — decisions with calibrated probability distributions — which is what makes the confidence field of a choice or score answer a trustworthy input to routing policy rather than a stylistic flourish [TypeSafe AI, 2026e,b]. The distinction matters practically: confidence-gated routing (4) is only sound if the confidence numbers are calibrated, which is a training-time property, not a prompt-time one.

**Agent decision layers.** Where agent frameworks place the model *in* the control loop — the model chooses its next action — the System One placement is the inverse: code owns control flow and the model supplies atomic judgments at fixed points [TypeSafe AI, 2026d]. Third-party projects in the Jev ecosystem explore this decision-layer space from adjacent angles: an architectural registry for composing Jev-style components [Claburn, 2026], a router that orchestrates decision calls across application workflows [Corvin, 2026], and a standalone evaluation harness for Jev question sets [Morales, 2026]. `daf-jev` is complementary rather than competitive: it contributes a dependency-light, strictly typed *client and composition library* — the tested plumbing layer these efforts can sit on — together with measured evidence for the batching and confidence-routing patterns the ecosystem’s documentation describes qualitatively [TypeSafe AI, 2026c].

## 10 References

The bibliography lives in `manuscript/references.bib` and is resolved by Pandoc — `--natbib` on the PDF path, `--citeproc` for the other editions. Citation keys used throughout this manuscript are fixed by the project’s research pass: the primary System One source [TypeSafe AI, 2026d], its concept primer [TypeSafe AI, 2026e], the API reference [TypeSafe AI, 2026a], the confidence documentation [TypeSafe AI, 2026b], the decision patterns documentation [TypeSafe AI, 2026c], and the third-party Jev ecosystem works [Claburn, 2026], [Corvin, 2026], and [Morales, 2026]. Every bracketed key in the preceding sections resolves against that file. Further background sources are available in `references.bib`.

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